

Tensor Analysis

3. Contravariant and Covariant Vectors

Suppose N quantities $A^1, A^2, \dots, A^N$ in a coordinate system $(x^1, x^2, \dots, x^N)$ are related to N other quantities $\bar{A}^1, \bar{A}^2, \dots, \bar{A}^N$ in another coordinate system $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N)$ by the transformation equations

$$\bar{A}^p = \sum_{q=1}^N \frac{\partial x^p}{\partial \bar{x}^q} A^q \quad p = 1, 2, \dots, N$$

which by the conventions adopted can simply be written as

$$\vec{\bar{A}}^p = \frac{\partial x^p}{\partial \bar{x}^q} A^q$$

Then they are called components of a contravariant vector or contravariant tensor of the first rank or first order.

On the other hand, suppose N quantities $A_1, A_2, \dots, A_N$ in a coordinate system $(x^1, x^2, \dots, x^N)$ are related to N other quantities $\bar{A}_1, \bar{A}_2, \dots, \bar{A}_N$ in another coordinate system $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N)$ by the transformation equations

$$\bar{A}_p = \sum_{q=1}^N \frac{\partial x^q}{\partial \bar{x}^p} A_q \quad p = 1, 2, \dots, N$$

Or
$$\bar{A}_p = \frac{\partial x^q}{\partial \bar{x}^p} A_q$$

Then they are called components of a covariant vector or covariant tensor of the first rank or first order.

Note that a superscript is used to indicate contravariant components whereas a subscript is used to indicate covariant components; an exception occurs in the notation for coordinates.

Instead of speaking of a tensor whose components are A^p or A_p , we shall often refer simply to the tensor A^p or A_p .